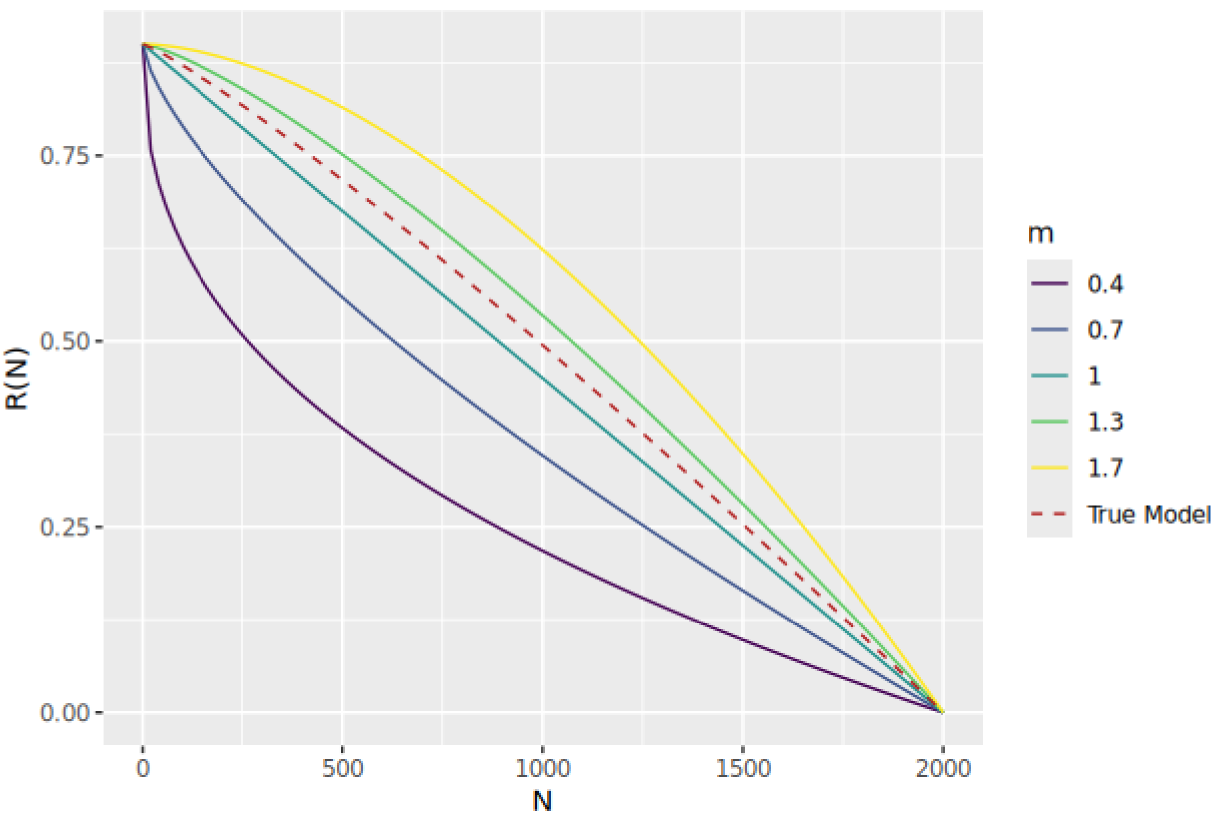
Coypu adaptive management

We will use the following discrete-time population model, where r is the intrinsic rate of growth, K is the carrying capacity, $q(u)$ is the harvest rate as a function of action u, R is the per-capita growth rate, and m whether the per-capita growth rate is linear, concave, or convex:

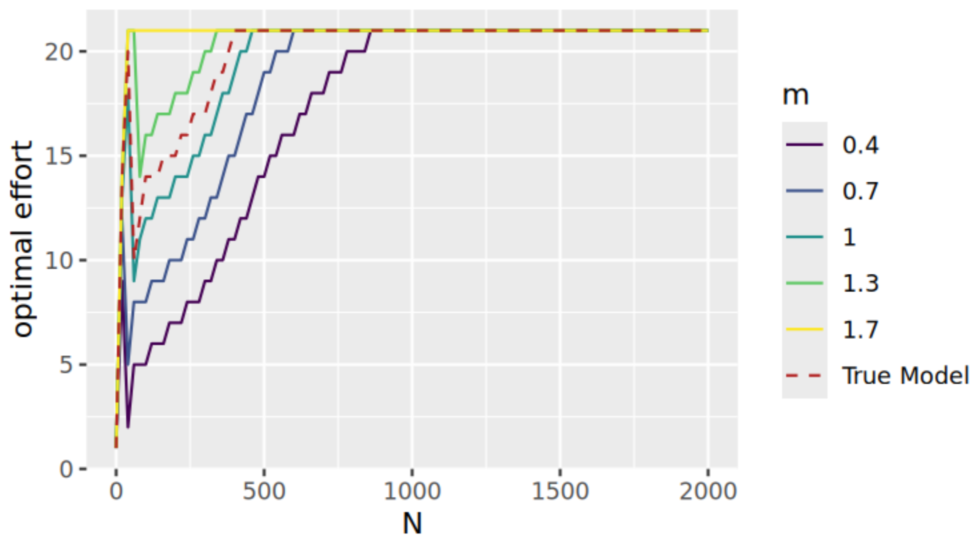
$$N_{t+1} = N_t(1 + R(N_t)) - q(u) \times N + \epsilon$$
$$R(N_t) = r \times (1 - (N_t/K)^m)$$

If $m > 1$, most density—dependent change occurs at high population levels (close to the carrying capacity), which is theoretically common for species with life history strategies typical of large mammals. If $m < 1$, most density-dependent change occurs at low population levels, which is theoretically common for species with life history strategies typical of insects and some fishes (Fowler, 1981).

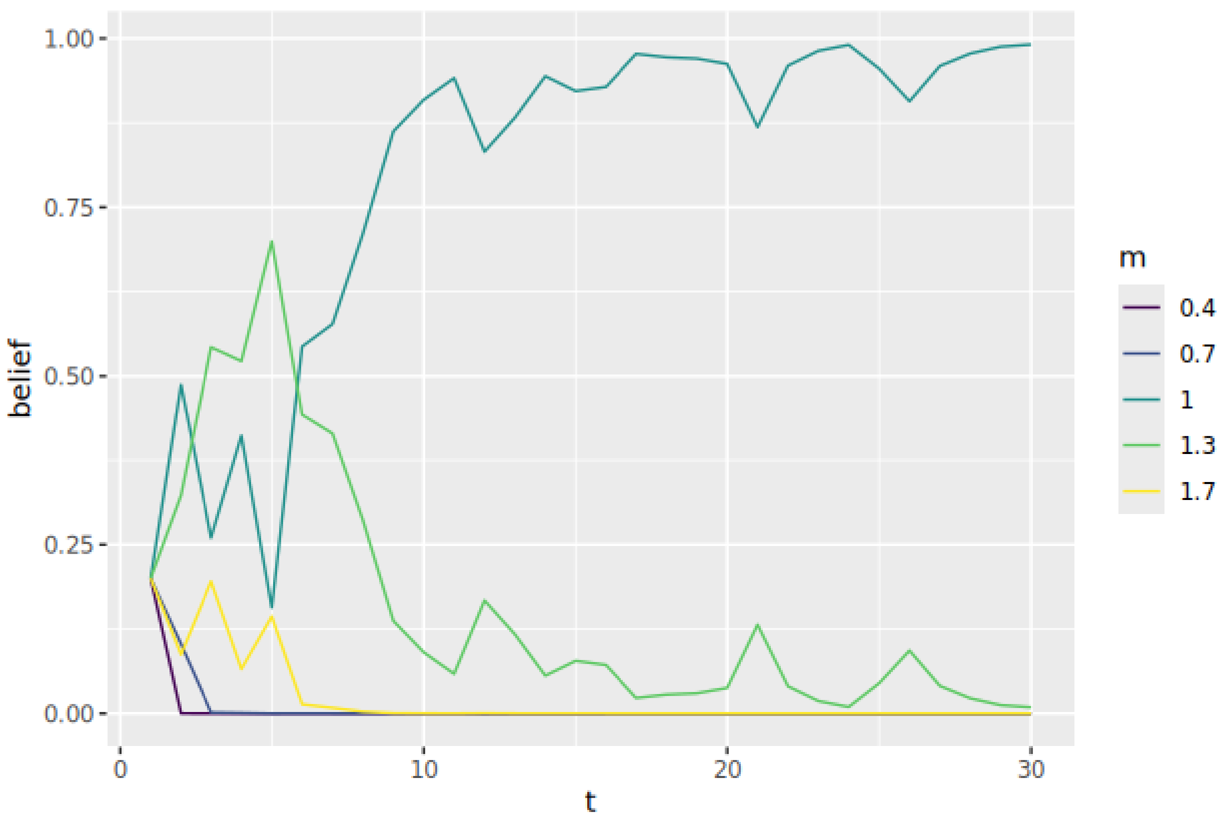
We will consider the following model set, with five possible values of m. This model bounds, but does not contain, the true model we will use to simulate dynamics.



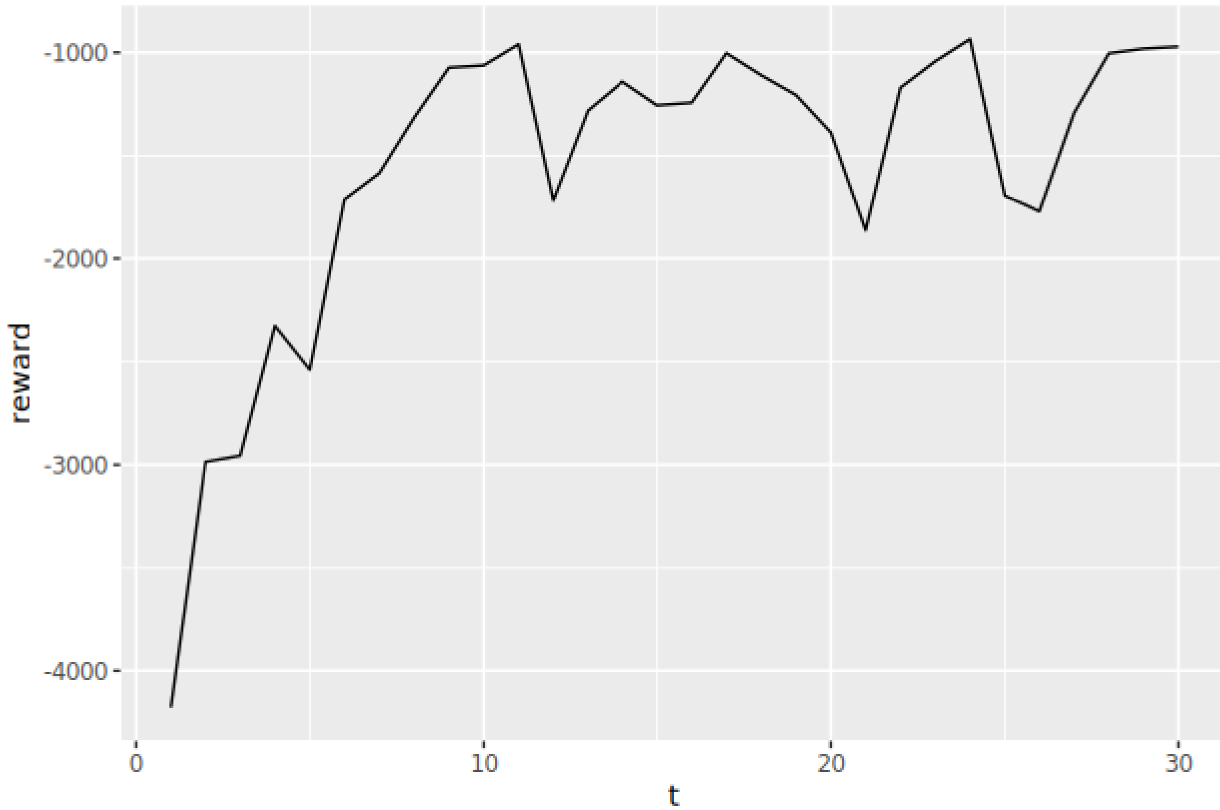
These different values of m result in different optimal policies (with known dynamics):



We will use adaptive management, which will update the belief state in the five models over time:



And we will see an increase in reward over time:



References

Fowler, C. W. (1981). Density dependence as related to life history strategy. Ecology, 62(3), 602-610.